



Universität
Zürich^{UZH}

Institute of Psychology

Reporting Results from Cognitive Models

Satellite Event: Analyzing Data on the Level of Cognitive Processes

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Workshop: Overview

Monday

Morning

Intro
Why
hierarchical
models?

Signal-
Detection
Models
(brms)

Tuesday

Diffusion
Decision
Model
(bmm)

Practical
Exercises

Materials are
available via
GitHub

30 Min. Coffee
Breaks
10:30 & 15:00
60 Min. Lunch
Break
12:30

Afternoon

Memory
Measurement
Model
(bmm)

Practical
Exercises

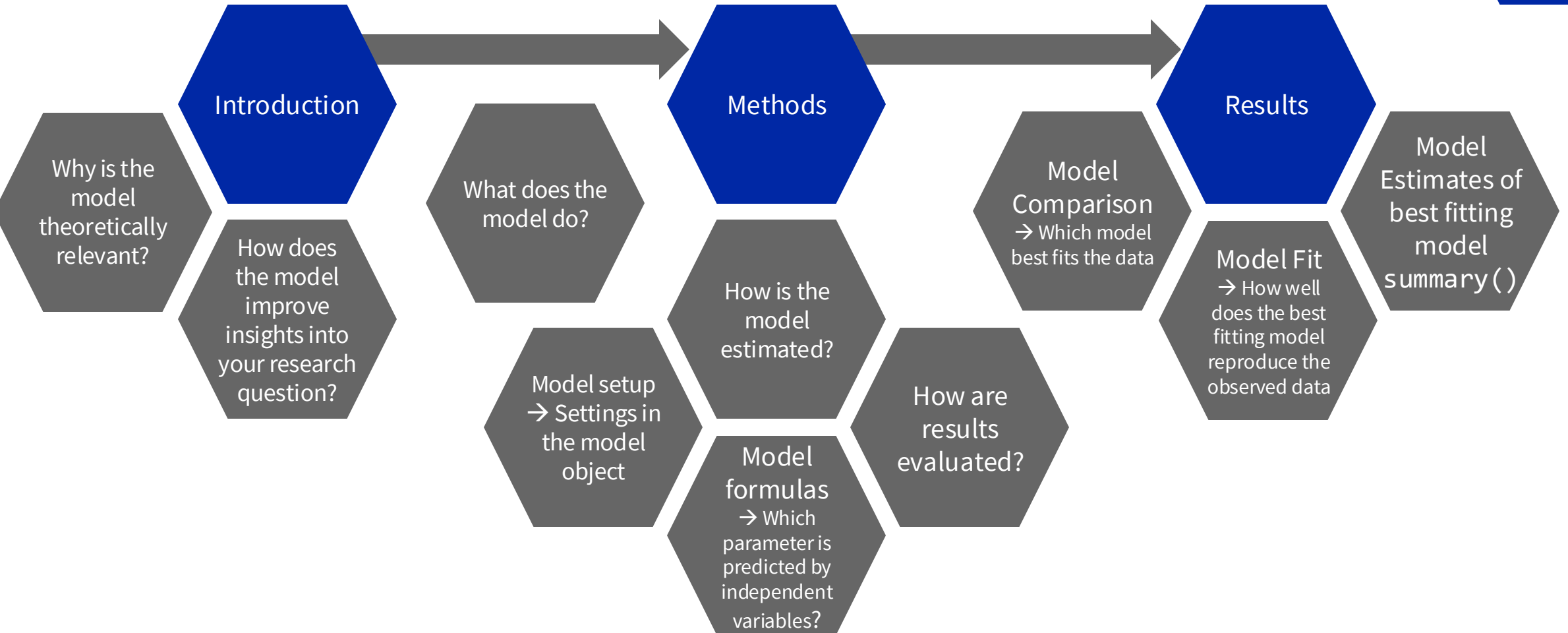
Signal
Discrimination
Model

Reporting
Results from
Cognitive
Modelling
Analyses

Ask questions
whenever
something is
not clear
😊

Reporting Results from Cognitive Models

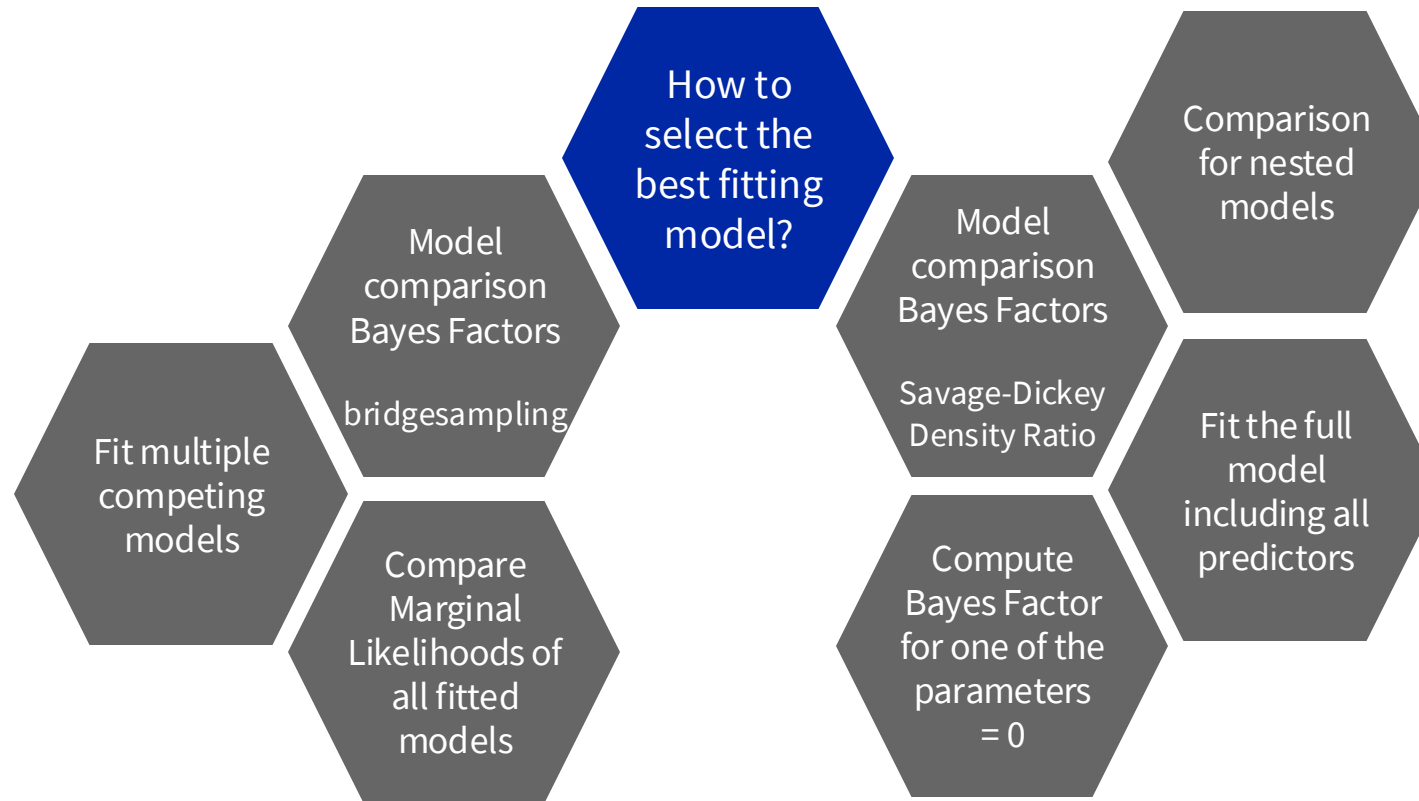
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Model selection

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Model comparisons: Bridgesampling

- Bayes Factor: ratio of two marginal likelihoods

$$P(H|D) = \frac{P(H) * P(D|H)}{P(D)}$$

Marginal
Likelihood
= probability
of the data

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$$P(D) = P(H) * P(D|H) + P(\neg H) * P(D|\neg H)$$

$$BF = \frac{\text{Marginal likelihood of Model 0}}{\text{Marginal likelihood of Model 1}}$$

$$= \frac{P(H, \text{Model 0}) * P(D|H, \text{Model 0}) + P(\neg H, \text{Model 0}) * P(D|\neg H, \text{Model 0})}{P(H, \text{Model 1}) * P(D|H, \text{Model 1}) + P(\neg H, \text{Model 1}) * P(D|\neg H, \text{Models})}$$

$$= \frac{P(D|\text{Model 0})}{P(D|\text{Model 1})}$$

Model comparisons: Bridgesampling

Estimating Bayes Factors via bridgesampling:

- In the R package *bridgesampling* you can estimate the marginal likelihood:

$BF_{01} = \text{bridgesampling::bayes_factor}(\text{model } 0, \text{model } 1)$

```
model1 = brms::brm(  
  formula = y ~ 1,  
  data = data,  
  cores = 4,  
  warmup = 1000,  
  iter = 13500,  
  chains = 4,  
  save_pars = brms::save_pars(all = TRUE)  
)
```

```
model2 = brms::brm(  
  formula = y ~ 1 + group_numeric,  
  data = data,  
  cores = 4,  
  warmup = 1000,  
  iter = 13500,  
  chains = 4,  
  save_pars = brms::save_pars(all = TRUE)  
)
```

Important:

save_pars argument in
brm function has to be set to
save_pars(all = TRUE) for the
bridgesampling to work

Hypothesis tests: Savage-Dickey Density Ratio

„Is parameter θ different from 0?“ → nested models

$$BF = \frac{P(D|H_0)}{P(D|H_1)} \quad H_0: \theta = 0, H_1: \theta \neq 0$$

Instead of model comparisons, we can compare parameter values of interest (e.g., 0)

The most common way of calculating Bayes Factors for parameters is the:

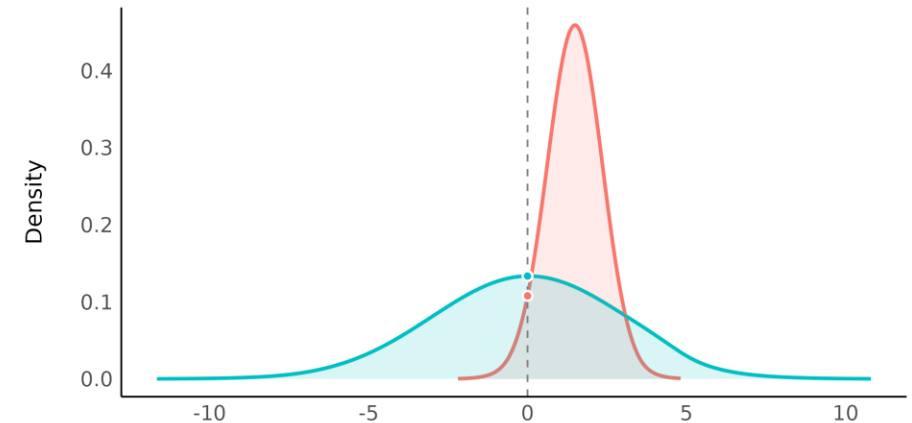
Savage-Dickey Density Ratio (SDDR)

= the ratio of the prior to the posterior at a certain parameter value

How did my belief in a certain parameter value (e.g., 0) change before and after having seen the data?

$$BF_{01} = \frac{P(\theta = 0 | \text{Posterior})}{P(\theta = 0 | \text{Prior})}$$

approximation of the BF comparing a model that includes the parameter of interest to a model in which the parameter is fixed at parameter value of interest (e.g., 0)



Distribution ■ posterior ■ prior

Hypothesis tests: Savage-Dickey Density Ratio

In the `brms` package you can calculate the SDDR using `hypothesis()`:

```
hypothesis(fit, "parameter = 0")
```

```
> hypo_hebb
Hypothesis Tests for class b:
      Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio Post.Prob Star
1 (block:hebb_trial... = 0      0.17      0.02    0.14    0.2      0          0    *
---
'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
'*': For one-sided hypotheses, the posterior probability exceeds 95%;
for two-sided hypotheses, the value tested against lies outside the 95%-CI.
Posterior probabilities of point hypotheses assume equal prior probabilities.
```

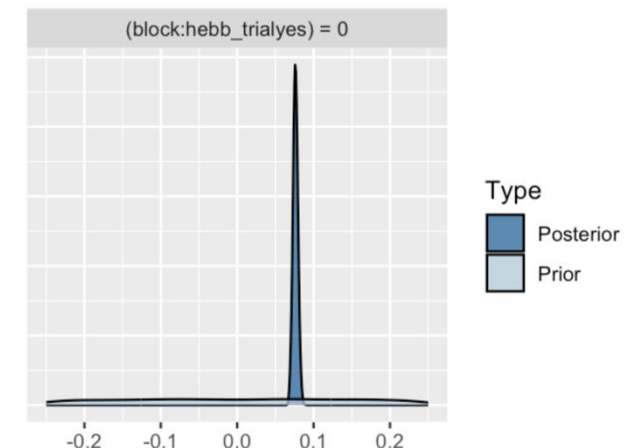
Important:

- 1) You need to specify priors for the parameters of interest
- 2) To get the Evid.Ratio (= BF) you need to set the `sample_prior` option in `brm()` to „yes“

```
mymodel <- brm(myformula,
  data = mydata,
  family = binomial(link = logit),
  prior = myprior,
  chains = n_chains,
  warmup = n_warmups,
  iter = n_warmups + n_iters,
  sample_prior = "yes",
  cores = getOption("mc.cores", n_chains),
  file = here("Models", "brms_model_sddr"),
  file_refit = "on_change")
```

If you save your hypothesis in an object, you can also plot the prior against the posterior:

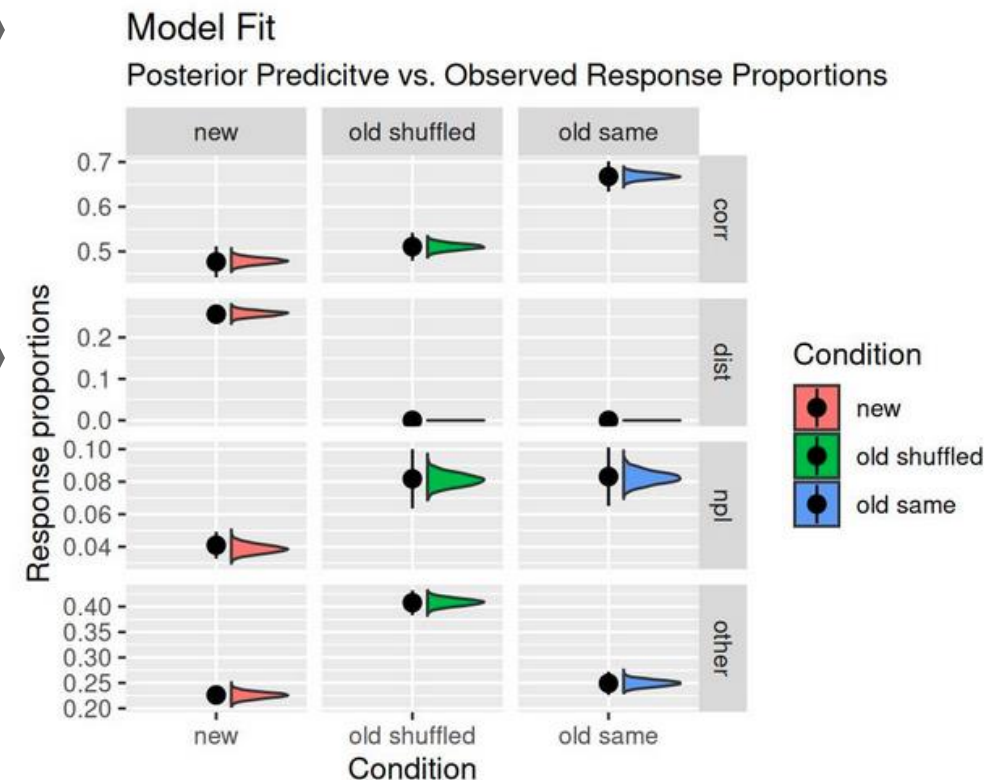
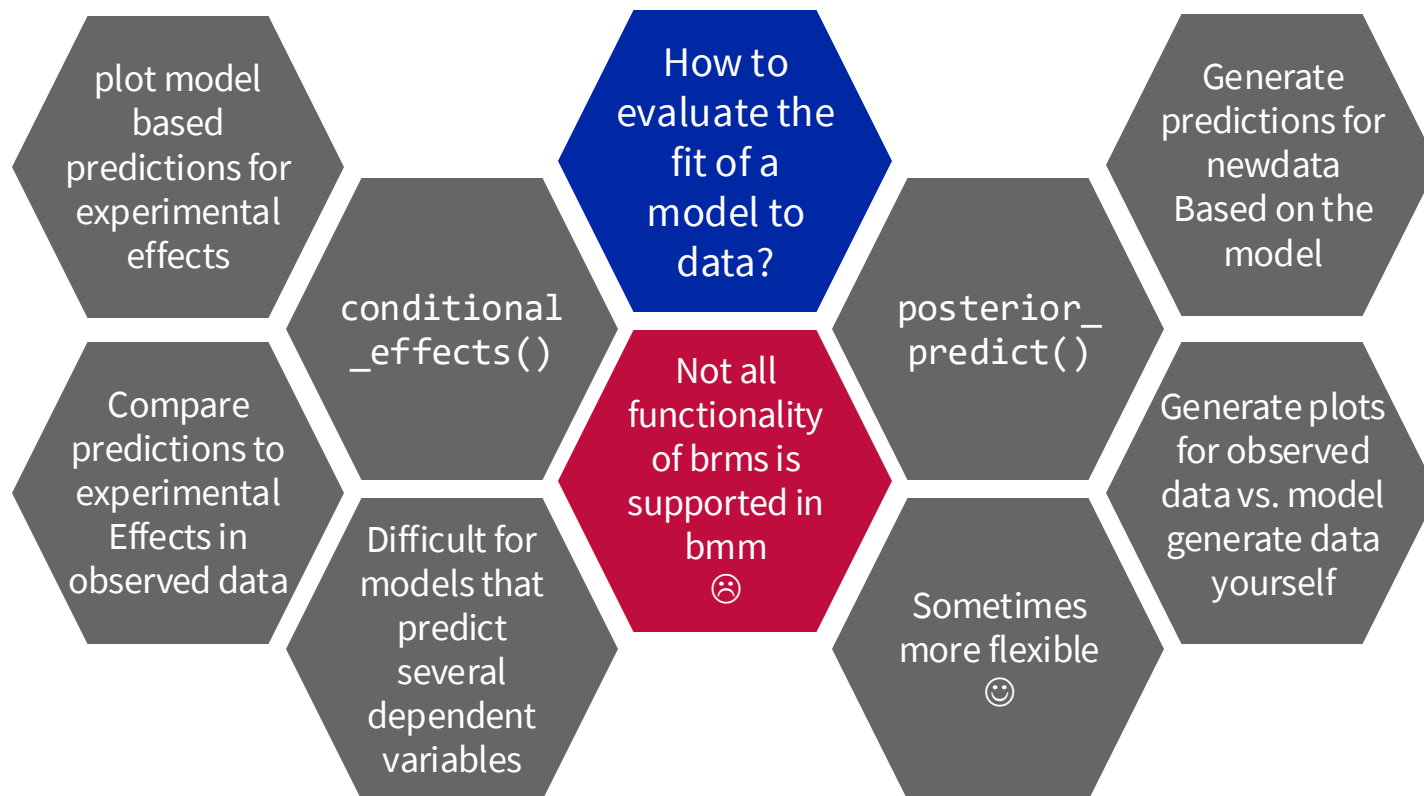
```
plot(hypothesis)
```



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Model fit

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What are your questions so far?

